

Design, Instrumentation & CFD Validation of a Miniature Open-Circuit Subsonic Wind Tunnel for Low-Reynolds-Number Aerodynamics

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GROUP
A13

Motivation & objectives

Low-Reynolds-number aerodynamic data is scarce for UAVs and ground vehicles. We built a modular, low-cost tunnel for rapid physical validation of airfoils and bluff bodies.

- Design and fabricate a subsonic wind tunnel
- Measure lift, drag and surface pressure on two models
- Validate the measurements against Ansys Fluent, including the model's mounting stand
- Visualize flow attachment and separation

Tunnel design



Fig. 1 – Assembled tunnel with the Ahmed body model mounted in the test section

Flow path: settling chamber with honeycomb → 6.25:1 contraction → test section → diffuser → fan. Laser-cut plywood + 3D-printed contraction.

160×160×340

mm test section

6.25 : 1

contraction (400×400 → 160×160 mm)

57–65 km/hr

test-section wind-speed

Re ≈ 1.2×10⁵

based on 100 mm chord

Honeycomb L/d ≈ 7

70 mm straw honeycomb

350 W

14 in axial fan, 2800 RPM, 2000+ CFM

Instrumentation



Air speed

Pitot tube + MPXV7002DP differential pressure sensor, zero-tared each run



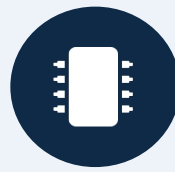
Lift & drag

Dual-lever mechanical amplification (4:1 lift, 4.5:1 drag), two 1 kg load cells + HX711, ball-bearing pivots



Surface pressure

Taps → silicone tubing → same MPXV7002DP, referenced to Pitot static; switched by hand, one at a time



Data

Arduino Mega → MATLAB for C_p, drag polars and L/D

Test models

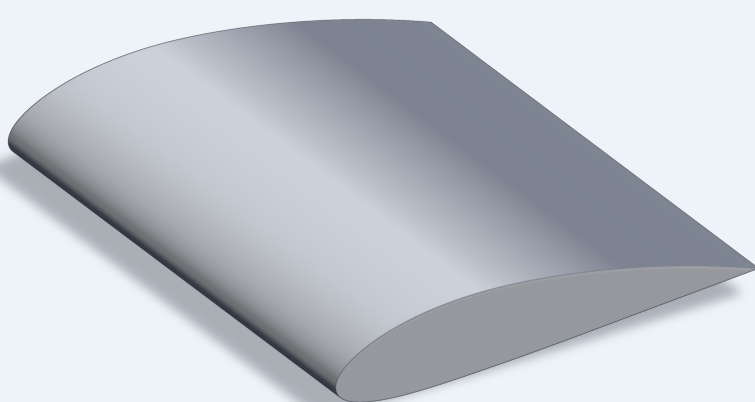


Fig. 2a – NACA 4418 airfoil

NACA 4418 airfoil

100 mm chord, 125 mm span
4 upper + 3 lower surface pressure taps
Ref. area 0.0125 m²

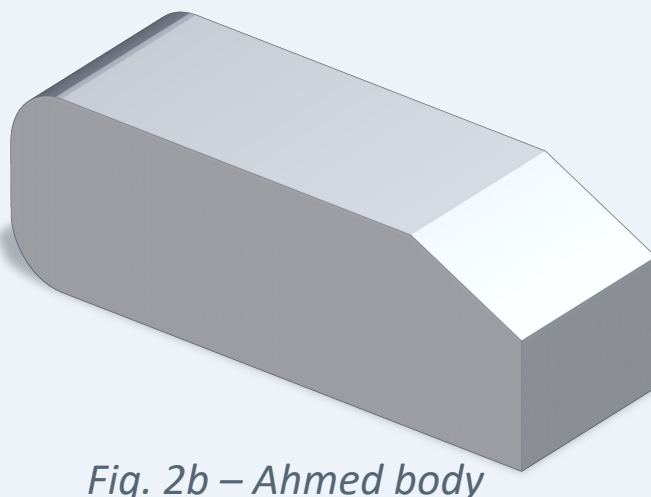


Fig. 2b – Ahmed body

Modified Ahmed body

120 × 40 × 37 mm, 25° rear slant
8 upper-surface pressure taps
Ref. area 1.48×10⁻³ m²

CFD setup (Ansys Fluent)

- Steady RANS, k- ω SST; Fluent 2026 R1 Student
- Real test-section dimensions with “no-slip” tunnel walls, and the model stand included in one run, excluded in another
- 15–18 prism layers, $y^+ \approx 1.7$; oscillating cases averaged over limit cycles
- Coefficients (C_l, C_d, C_p) compared, not raw forces/pressures

Results: pressure & force

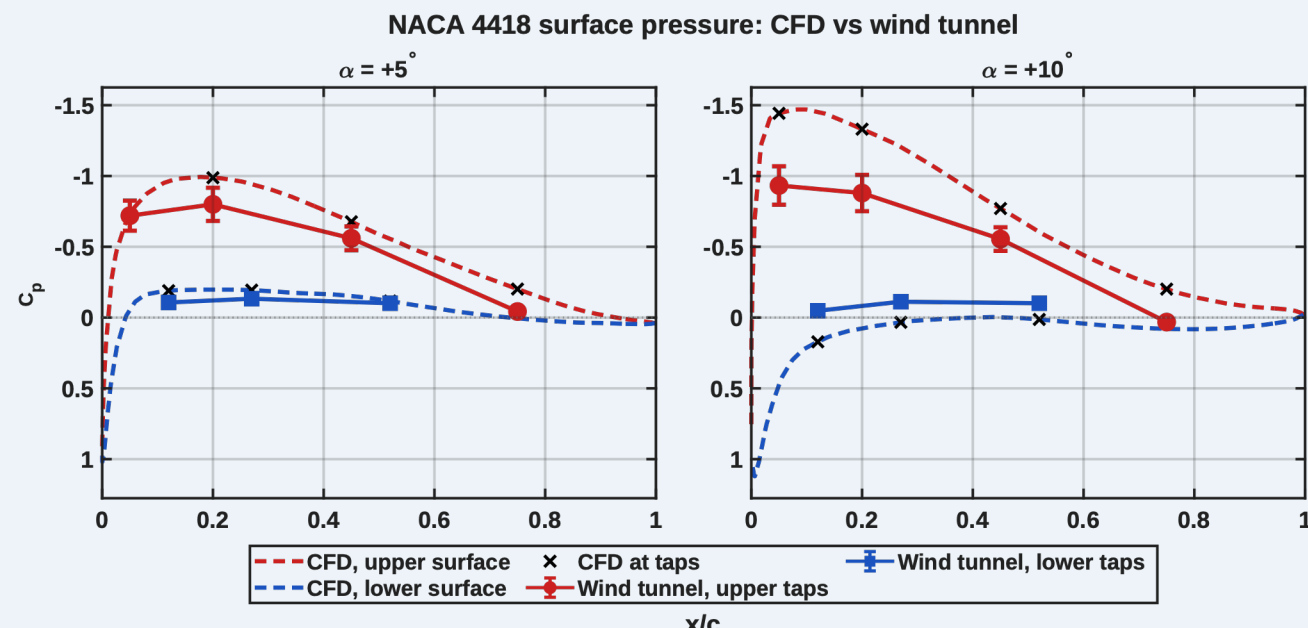


Fig. 3a – NACA 4418 surface C_p vs x/c, WT vs CFD at 5° and 10° AoA

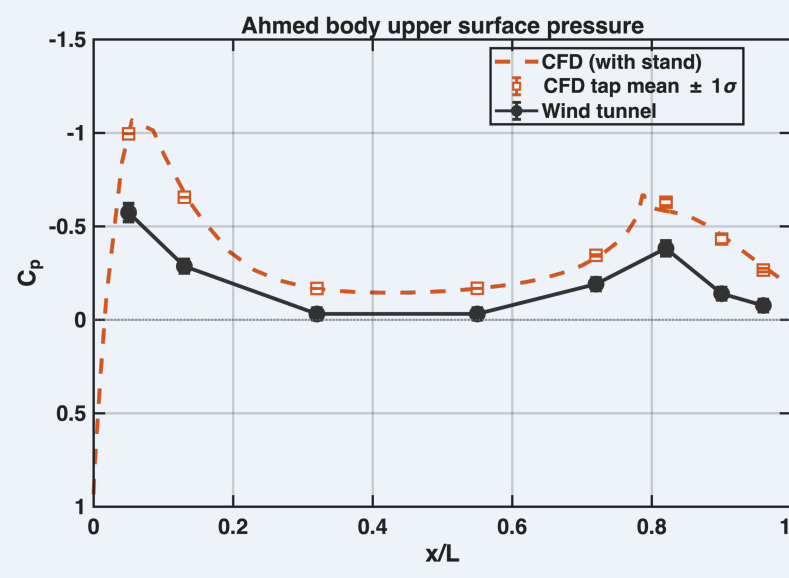


Fig. 3b – Ahmed body surface C_p along its length, WT vs CFD

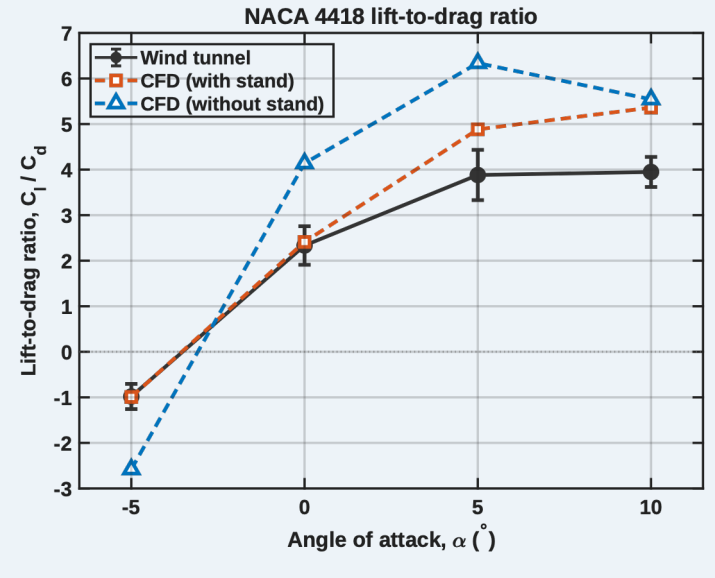


Fig. 3c – NACA 4418 L/D vs AoA — WT vs CFD (with stand) vs CFD (no stand)

CFD validation

Case	Coeff.	Exp.	CFD	Δ vs CFD
NACA 0°	C _l	0.176	0.177	-0.7%
	C _d	0.075	0.073	+2.6%
NACA +5°	C _l	0.354	0.417	-15.1%
	C _d	0.091	0.085	+6.8%
NACA +10°	C _l	0.612	0.762	-19.7%
	C _d	0.155	0.142	+9.1%
NACA -5°	C _l	-0.062	-0.072	+13.0%
	C _d	0.064	0.072	-12.0%
Ahmed body	C _l	0.671	0.616	+9.0%
	C _d			

Δ = (Exp - CFD)/|CFD|. Green: within 10%. CFD includes the model stand. At 5° and 10° the lift gap is 2.6–4× the uncertainty: low-Re boundary-layer effects SST misses, effective AoA under load and balance calibration.

≤ 3%

lift and drag agreement with CFD at 0° AoA once the stand is modeled in CFD. Errors increase considerably with higher AoA

With stand-free CFD: drag +80%

Flow visualization (Qualitative Analysis)



Fig. 4a – NACA 4418, -5° AoA. Glycerin smoke rake.

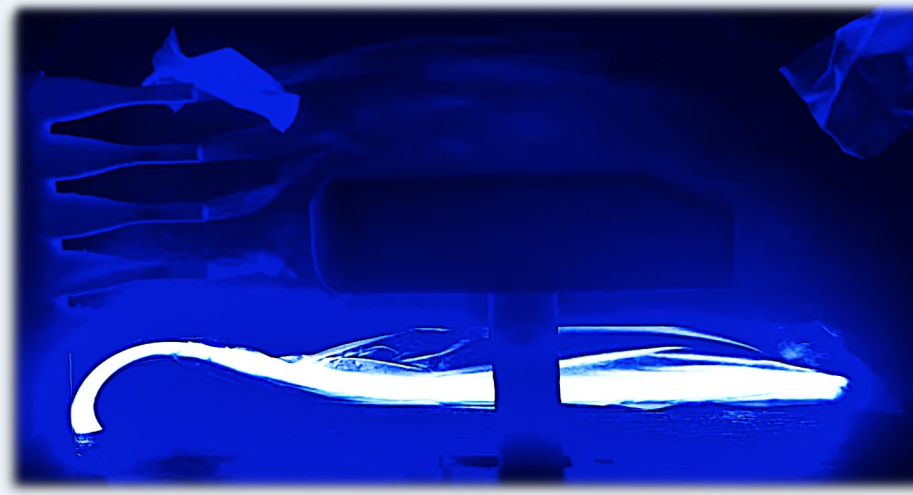


Fig. 4b – Ahmed body (preliminary). Older smoke system; not yet re-run on the upgraded rig.

- Glycerin fog/smoke rake shows attachment, separation and wake structure
- Used qualitatively alongside the C_p and force data
- NACA: attached on both surfaces at 0°; at -5° the lower-surface smoke turns diffuse
- Ahmed body: preliminary run only; its 25° slant is below the ~30° critical angle

Key conclusions

- **Tunnel:** 16–18 m/s (Re ≈ 1.2×10⁵) in a 160×160 mm test section behind a 6.25:1 contraction
- **NACA @ 0° AoA:** with the stand modeled, lift and drag agree with CFD within 3%; without the stand in CFD the gap grows to +80%
- **NACA @ +5°, +10° AoA:** measured lift 15–20% below CFD: low-Re boundary layer, effective AoA under load, balance calibration
- **NACA Pressure:** C_p shapes and suction peaks match CFD; suction is weaker, mainly from pressure-line leakage
- **Ahmed body:** drag 9% above CFD (within uncertainty). C_p reproduces every CFD feature, with weaker suction (≈ 0.65 × CFD).

Future work

- Stand-only tare run to isolate model drag
- Blockage correction (~10%)
- 16-bit ADC (e.g. ADS1115), more taps
- Repeatable AoA setting; refined smoke rake
- Model-mounted, combined-load calibration

Acknowledgments

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References

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- [3] J. B. Barlow, W. H. Rae, A. Pope, Low-Speed Wind Tunnel Testing, 3rd ed., Wiley, 1999.